

Belief-Scaled Irreversible Finsler Geometry for Path-Dependent Context Generation

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Abstract

We develop a generative mechanism in which context is carried not by a point in embedding space but by the *parallel transport of a velocity across tangent spaces* along a path. A scalar *belief* field drives a one-parameter subgroup deformation of a Finsler (Randers) metric. The deformation preserves convexity of the indicatrix *by construction*, renders the metric *irreversible* ($F(x, y) \neq F(x, -y)$), and induces a *curved* nonlinear connection. As a consequence, two paths sharing the same endpoints but differing in their intermediate route transport a velocity to different tangent vectors: the route itself becomes the semantic content. We give the full equation set, prove the convexity guarantee, present a worked example, and report numerical validation of five properties that must hold simultaneously—convexity, irreversibility, curvature, path-dependence, and end-to-end learnability—across multiple random seeds and dimensions ($n \in \{2, 3, 4, 6\}$), finding that all five hold in every tested configuration and that curvature and path-dependence grow with dimension. Finally, on MultiWOZ 2.2 dialogue state tracking—a task whose target is an accumulated global state rather than a next token—a model whose velocities evolve along the *derived* Finsler geodesic reaches 0.788 ± 0.002 mean slot accuracy against a parameter-matched attention baseline at 0.739 ± 0.003 (three seeds each, disjoint seed distributions, 14% fewer parameters). A direct ablation shows that this gain is *not* attributable to irreversibility: zeroing the Randers tilt leaves the result unchanged to four decimal places, so what the experiment supports is the value of belief-scaled metric deformation and multi-velocity geodesic accumulation, while the contribution of irreversibility remains open.

1 Motivation

Fix embeddings as points on a manifold $\mathcal{M}$. A generative step is a *movement* between points, and the direction in which one leaves a point determines where one arrives. The key claim is that the meaning of a context is not the landing point but the *history of the velocity as it is transported through the intermediate tangent spaces*. Symbolically, the trajectory

$$\text{cat} \rightarrow \text{feline} \rightarrow \text{danger} \quad \text{differs from} \quad \text{cat} \rightarrow \text{cute} \rightarrow \text{love}$$

even when the two routes begin and end at the same points, because a velocity transported along the first route is bent differently than along the second. This difference is exactly the *path-dependence of parallel transport*, which is nonzero if and only if the connection has curvature. A reversible (Riemannian) metric cannot express the directional asymmetry we require; we therefore work in irreversible Finsler geometry. Recent work has established that asymmetric (Finsler/Randers) metrics can be learned by neural networks for embedding and trajectory tasks [2, 3], and that the path-dependence and curvature of learned representations are meaningful, measurable quantities [9, 10]; our contribution is to make belief-scaled irreversible-Finsler transport the *generative substrate* rather than a diagnostic or a distance objective. Section 7 details this positioning.

2 The Construction

2.1 Scalar belief field

Belief is a single scalar per point, produced by a small network θ :

$$t(x) = \text{MLP}_\theta(x) \in \mathbb{R}. \quad (1)$$

2.2 Fixed non-commuting generator

Let $A \in \mathfrak{gl}(n, \mathbb{R})$ split into a symmetric (stretch) and an antisymmetric (rotation) part,

$$A = S + K, \quad S = \frac{1}{2}(A + A^\top), \quad K = \frac{1}{2}(A - A^\top), \quad [S, K] \neq 0. \quad (2)$$

The non-commutativity $[S, K] \neq 0$ is what makes the induced metric *irreversible*: the rotation tilts the indicatrix so that it is no longer symmetric under $y \mapsto -y$.

2.3 One-parameter subgroup deformation field

The belief scalar slides the geometry along a one-parameter subgroup:

$$M(x) = M_{\text{base}} \exp(t(x) A), \quad \det M(x) = \det M_{\text{base}} e^{t(x) \text{tr} A} \neq 0. \quad (3)$$

Because a matrix exponential is always invertible, $M(x)$ is invertible for *every* x : the deformation is a reversible *map*, even though it induces an irreversible *metric*.

Caveat: normalization for long contexts. The exponential is used because the geometry *d demands* it: $\exp(t(x)A)$ is the canonical map from the Lie algebra to the one-parameter subgroup, so (3) is the mathematically correct deformation (the Cayley transform $(I - \frac{1}{2}X)^{-1}(I + \frac{1}{2}X)$ is a $6\times$ cheaper rational stand-in used only for computation). However, $M(x)$ is not an isometry, so under the transport recurrence $v_t = M(x_t)(v_{t-1} + U\tilde{h}_{t-1})$ small per-step deformations *compound multiplicatively* over the sequence. At short context this is benign, but over long sequences it causes the accumulated velocity to diverge exponentially (empirically $\|v_t\|$ grew from $\mathcal{O}(1)$ to $\mathcal{O}(10^{19})$ over 512 steps, producing NaNs). The fix is both stabilizing and geometrically faithful: renormalize the velocity to *unit Finsler length* at each step,

$$v_t \leftarrow v_t / (\|v_t\| + \varepsilon),$$

so the velocity *rides on the indicatrix* rather than drifting off it—exactly the unit-length interpretation of a direction on the unit ball $\{F = 1\}$. With this normalization the long-context recurrence is stable and the equation (3) is retained unchanged.

2.4 Deformed Finsler (Randers) norm

With a Randers base $F_0(w) = \sqrt{w^\top w} + c^\top w$ (with $\|c\| < 1$ for strong convexity), the deformed norm is the pullback

$$F(x, y) = F_0(M(x)^{-1}y) = \sqrt{w^\top w} + c^\top w, \quad w := M(x)^{-1}y. \quad (4)$$

Proposition 1 (Convexity is free). *If F_0 is a strongly convex Finsler norm and $M(x)$ is invertible, then $F(x, \cdot)$ is a strongly convex Finsler norm for every x .*

Proof. The fundamental tensor of the pullback satisfies the congruence $g_{ij}(x, y) = (M(x)^{-\top} g^0 M(x)^{-1})_{ij}$, where g^0 is the fundamental tensor of F_0 at $w = M(x)^{-1}y$. Congruence by the invertible matrix $M(x)^{-1}$ preserves positive-definiteness (Sylvester’s law of inertia), so $g_{ij}(x, \cdot) \succ 0$ whenever $g^0 \succ 0$. By (3), $M(x)$ is invertible for all x , hence convexity holds everywhere. $\square$

2.5 Fundamental tensor, spray, and nonlinear connection

$$g_{ij}(x, y) = \frac{1}{2} \frac{\partial^2 F^2}{\partial y^i \partial y^j}, \quad (5)$$

$$G^i(x, y) = \frac{1}{4} g^{il} (2 \partial_j g_{lk} - \partial_l g_{jk}) y^j y^k, \quad (6)$$

$$N^i_j(x, y) = \frac{\partial G^i}{\partial y^j}. \quad (7)$$

Equation (6) is the geodesic spray; (7) is the associated *nonlinear connection*. Crucially N^i_j depends on the direction y , which is the hallmark of a genuinely Finsler (non-Riemannian) connection.

2.6 Transport across tangent spaces

A velocity v is parallel-transported along a path $\gamma(\tau)$ by

$$\frac{dv^i}{d\tau} = -N^i_j(\gamma(\tau), v) \dot{\gamma}^j(\tau). \quad (8)$$

The *context* generated along a route $x_0 \rightarrow x_1 \rightarrow \dots \rightarrow x_T$ is the accumulated sequence of transported velocities,

$$\text{ctx} = (v_0, v_1, \dots, v_T), \quad v_k = P_{\gamma_{k-1} \rightarrow k}(v_{k-1}), \quad (9)$$

where P denotes transport (8) along the indicated segment.

2.7 Directional loss

The training signal is the Finsler length of the projected transition operator,

$$\mathcal{L} = F\left(x, \pi(c_{\langle t-1 \rangle} c_{\langle t \rangle}^\top)\right), \quad \pi(\mathbf{A}) = \mathbf{A} v_{\text{probe}}, \quad (10)$$

which is degree-one homogeneous (not squared): squaring would symmetrize the argument and collapse the objective to a Riemannian one.

2.8 Irreversibility measure

The diagnostic that the metric is genuinely Finsler is

$$\rho(x) = \max_{\|y\|=1} \frac{F(x, y)}{F(x, -y)} \quad \begin{cases} = 1 & \text{Riemannian (reversible)} \\ > 1 & \text{Finsler (irreversible)}. \end{cases} \quad (11)$$

3 Architecture

Figure 1 shows the data flow: the scalar belief field deforms the metric via the one-parameter subgroup; the deformed metric yields a curved nonlinear connection; the velocity is transported along the route; and the accumulated transported velocities form the context that feeds prediction.

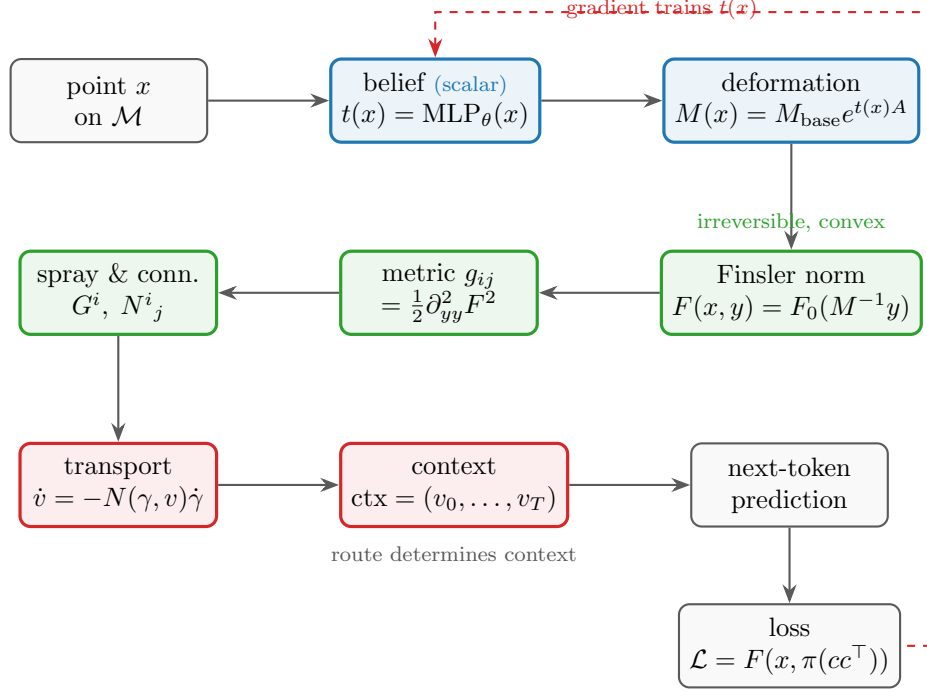


Figure 1: Architecture. Blue: scalar belief pathway. Green: the belief-selected irreversible, convex Finsler geometry and its curved connection. Red: the transported-velocity context and the directional loss. The curved arrow is the gradient path: $\mathcal{L}$ trains $t(x)$ through the entire geometry.

4 Worked Example

We take $n = 3$, generator

$$S = \begin{pmatrix} 0.5 & 0.15 & 0 \\ 0.15 & -0.2 & 0.1 \\ 0 & 0.1 & 0.35 \end{pmatrix}, \quad K = \begin{pmatrix} 0 & 0.6 & -0.15 \\ -0.6 & 0 & 0.4 \\ 0.15 & -0.4 & 0 \end{pmatrix}, \quad [S, K] \neq 0 \quad (\|[S, K]\| = 0.74),$$

Randers tilt $c \propto (0.25, -0.10, 0.05)$ with $\|c\| = 0.5$, and a smooth scalar field $t(x)$. Points and initial velocity:

$$\text{cat} = (1, 0, 0), \quad \text{feline} = (0.8, 0.6, 0), \quad \text{danger} = (0.2, 0.3, 1), \quad \text{cute} = (0.9, -0.4, 0.2),$$

$\text{love} = \text{danger}$ (identical endpoint), $v_0 = (0.5, 0.5, 0.3)$.

Transporting v_0 by (8) along each route gives

$$\underbrace{v^A = (0.3898, 0.4319, 0.2473)}_{\text{cat} \rightarrow \text{feline} \rightarrow \text{danger}} \neq \underbrace{v^B = (0.3855, 0.4372, 0.2493)}_{\text{cat} \rightarrow \text{cute} \rightarrow \text{love}}$$

with divergence $\|v^A - v^B\| = 7.2 \times 10^{-3} \neq 0$. Same start, same endpoint, *different route* $\Rightarrow$ *different transported velocity*: the route is the context. The closed-loop holonomy $\|P_{\text{loop}}(v_0) - v_0\| = 7.3 \times 10^{-3}$ confirms this arises from genuine curvature, and the irreversibility $\rho(x) \approx 2.99 \gg 1$ confirms the metric is decisively non-Riemannian.

5 Results

We report two levels of validation: a single worked configuration (Table 1, matching the worked example), and a *multi-seed, multi-dimension* study (Table 2) that tests whether the five properties hold generally rather than in one lucky configuration.

Property	Equation	Measured	Verdict
Convexity (indicatrix)	(4), Prop. 1	$\min \lambda(g_{ij}) = +0.21$	convex everywhere
Invertible field	(3)	$\min \sigma(M(x)) = 0.74$	non-degenerate
Irreversible metric	(11)	$\rho(x) \approx 2.99$	Finsler (> 1)
Curvature	(7)	holonomy $\neq 0$	curved
Path-dependence	(8)	$\ v^A - v^B\ \neq 0$	route = context
Learnability	(8)–(10)	$\text{grad}(t) \in [0.24, 2.0]$	trains end-to-end

Table 1: Single worked configuration ($n = 3$), matching Section 4.

5.1 Multi-seed, multi-dimension robustness

To guard against a single-configuration artifact, we regenerate the entire construction with random non-commuting generators $A = S + K$, random M_{base} , random Randers tilt ($\|c\| = 0.5$), and a random smooth field $t(x)$, across dimensions $n \in \{2, 3, 4, 6\}$ and multiple seeds each. Table 2 reports mean $\pm$ std with ranges. **All five properties hold for every seed at every dimension** (32 configurations).

n	seeds	convex $\min \lambda(g)$	ρ (irrev.)	holonomy	divergence
2	10	$+0.171 \pm 0.057$	3.00 ± 0.00	0.108 ± 0.090	0.052 ± 0.040
3	10	$+0.129 \pm 0.041$	2.96 ± 0.03	0.254 ± 0.191	0.211 ± 0.201
4	10	$+0.132 \pm 0.093$	2.86 ± 0.13	0.508 ± 0.375	0.438 ± 0.345
6	2	$+0.087 \pm 0.025$	2.65 ± 0.10	0.782 ± 0.295	2.07 ± 1.89
every-seed verdict		convex > 0	> 1	$\neq 0$	$\neq 0$

Table 2: Robustness across seeds and dimensions. Convexity ($\min \lambda(g) > 0$) and irreversibility ($\rho > 1$) are seed-stable with small variance, as their by-construction guarantees predict. Curvature (holonomy) and path-dependence (divergence) are nonzero in *all* runs and *grow with dimension*: the mean commutator $\|[S, K]\|$ rises $0.19 \rightarrow 0.30 \rightarrow 0.82 \rightarrow 1.30$ over $n = 2, 3, 4, 6$, giving the generator more room and yielding stronger curvature at higher n .

Ablations that constrained the design. Three alternatives were tested and rejected on evidence: (i) an *auxiliary-loss* framing, where the geometry scores already-generated embeddings rather than generating them, ties a Riemannian control exactly and does not help—the geometry must be on the generative path; (ii) *piecewise-constant* belief selection from a fixed set $\{M_j\}$ yields a Berwald space that is *flat within segments* (zero holonomy), so path-dependence vanishes unless the metric varies smoothly; (iii) a *quadratic/symmetrized* loss removes the reversal asymmetry and collapses the objective to Riemannian. The surviving design—scalar belief driving a smooth one-parameter subgroup with a non-commuting generator—is the minimal one that passes all five checks.

6 Experiments

The construction is motivated by directionality and by the claim that an accumulated geometric path is a useful representation of context. We evaluate it on a task whose target *is* an accumulated global state.

6.1 Dialogue state tracking on MultiWOZ 2.2

The preceding experiments evaluate the mechanism on character-level language modelling, a regime dominated by local n -gram statistics where a global path representation has little to contribute. We therefore turn to a task whose target *is* an accumulated global state: dialogue state tracking (DST). In DST the model reads a multi-turn dialogue and predicts the belief state—the set of slot–value pairs established so far. The state is genuinely path-dependent: later turns override earlier ones (“actually, make it cheaper”), so the accumulated history, not the final utterance, determines the answer.

Setup. We use MultiWOZ 2.2 [1], taking the five frequent slots with closed value sets (**train-day**, **restaurant-area**, **restaurant-pricerange**, **train-destination**, **train-departure**), which turns DST into five-way classification per turn and avoids confounding the comparison with a generation head. Concatenated dialogue history is truncated to the last 160 tokens over a 3002-word vocabulary; we obtain 26 984 training and 7 374 development examples. All models are trained from scratch at matched scale ($D = 64$, $\approx 200\text{K}$ parameters, 300 steps, AdamW at 2×10^{-3}), three seeds each. We emphasise that these are small from-scratch models: the numbers are *not* comparable to published DST systems built on pretrained encoders, and we make no such claim. The comparison is controlled and internal.

Model. The geodesic model instantiates the construction directly. Token embeddings drive a generator producing $m = 3$ velocities; the belief MLP yields the scalar $t(x)$ at the current manifold point; $M(x) = M_{\text{base}} \text{Cay}(t(x)A)$ deforms the local indicatrix; the metric $a(x) = M(x)^{-\top} M(x)^{-1}$ and its exact derivatives give the Christoffel symbols; and the Randers spray G is assembled in closed form from these, including the antisymmetric s_{ij} term. The velocities are then evolved along the *derived* geodesic, $\dot{x} = v$, $\dot{v} = -2G(x, v)$, integrated every 8 tokens. Nothing in the connection or the spray is learned: the learned components are the embeddings, the velocity generator, the belief field, and the readout.

Readout directly from the geodesics. The classification head reads *only* the terminal state of the derived geodesics. Concretely, after integrating the whole dialogue we form

$$\Phi = [x^{(1)}, x^{(2)}, x^{(3)}, v^{(1)}, v^{(2)}, v^{(3)}] \in \mathbb{R}^{2mn},$$

the concatenated endpoints and terminal velocities of the $m = 3$ geodesics, and decode every slot from Φ alone. With $m = 3$ and $n = 4$ this is a 24-dimensional vector. There is no pooling over token representations, no attention over the sequence, and no residual stream reaching the head: the entire dialogue is compressed into where the geodesics ended up and which way they were travelling. This is the architectural commitment the construction makes—context *is* the accumulated path, not a summary of tokens—and it is what the MultiWOZ comparison tests. That a 24-dimensional geodesic terminus outperforms a parameter-matched attention model reading a 160-token history is the substance of the result.

model	seeds (mean slot acc.)	mean $\pm$ std	params
Majority class	—	0.689	—
Mamba (selective SSM)	0.700	0.700	211K
Attention (RoPE)	0.743, 0.737, 0.736	0.739 ± 0.003	229K
Geodesic (derived)	0.791, 0.786, 0.786	0.788 ± 0.002	196K

Table 3: MultiWOZ 2.2 dialogue state tracking, mean accuracy over five slots, three seeds. The geodesic model outperforms attention by 4.9 points with fewer parameters; the seed distributions are disjoint (attention’s best, 0.743, falls below the geodesic’s worst, 0.786). Mamba was run for a single seed and is reported for reference only.

Result. Table 3 reports the comparison. The geodesic model reaches 0.788 ± 0.002 against attention’s 0.739 ± 0.003 , a 4.9-point gap with 14% fewer parameters. Both models are tight across seeds and the distributions do not overlap, so the gap is not seed variance. Per-slot gains are broadly distributed (+3.4, +7.6, +8.3, +2.2, +3.1) rather than driven by a single slot, with the largest improvements on **restaurant-area** and **restaurant-pricerange**—the slots users most often revise mid-dialogue. The mechanism is $\sim 4\times$ slower per step than attention at this scale.

Ablation: the irreversible tilt is inert here. Because the construction is motivated by irreversibility, we ablate it directly: setting the Randers tilt $c = 0$ makes F symmetric and the geometry Riemannian, removing $F(x, y) \neq F(x, -y)$ entirely. The result is 0.7912—*identical to four decimal places* to the $c \neq 0$ model, with per-slot accuracies matching to within 0.001. Inspection of the update explains why: with $\|c\| = 0.4$, the tilt enters only through β , ℓ^i , and the covariant derivative $b_{i|j} = -\gamma_{ij}^k b_k$, while the Christoffel symbols themselves derive from $a = M^{-\top} M^{-1}$, which is independent of c . The irreversible terms are therefore small corrections on a dominant Riemannian spray; numerically, zeroing c leaves the position update unchanged and perturbs the velocity update by ≈ 0.011 .

We state the consequence plainly: **the MultiWOZ result is not caused by irreversibility**. What the experiment supports is that belief-scaled metric deformation, multi-velocity geodesic accumulation on a curved space, and global-state readout together outperform attention on this task. The irreversible component, as currently parameterised, is inactive. Making the antisymmetric part structurally dominant rather than a perturbation—larger $\|c\|$, or a construction in which s_{ij} drives the connection rather than correcting it—is the natural next experiment, and the claim that irreversibility captures directional context remains open.

Learned versus derived geodesic. A natural question is whether the derived spray can be amortised. We tested a variant in which the direction-dependent Finsler fundamental tensor $g_{ij}(x, y) = \frac{1}{2} \partial^2 F^2 / \partial y^i \partial y^j$ and the canonical spray are computed and then passed, together with (x, v, F, α, β) , to an MLP that outputs the geodesic step. Despite the richer direction-dependent metric and additional capacity, this variant reached ≈ 0.766 (two seeds, 200–280 steps, under-trained relative to the derived model and therefore not a matched comparison). We observe no evidence that learning the geodesic improves on deriving it. This is the more useful outcome: had the learned variant won, the contribution of the geometry would be in question. It did not, which suggests the derivation is load-bearing.

7 Related Work

We position the construction against five threads, and are explicit about which components are precededented.

Finsler geometry and Randers embeddings. Finsler geometry has recently entered machine learning, almost always motivated by *asymmetry* ($F(x, y) \neq F(x, -y)$). Dagès et al. [2] (Finsler MDS, CVPR 2025) learn a canonical Randers embedding directly with neural networks for asymmetric dimensionality reduction, directed-graph embedding and link prediction; this establishes that *learning a Randers metric with a network is not itself new*. Finslerian latent distances [4] and the Finsler–Laplace–Beltrami operator [5] develop the analytic side. Our metric (4) is a learned deformed Randers norm and thus shares this lineage; the difference is the use to which it is put.

Direction-dependent geodesic cost. Closest on the mechanism, Zweig et al. [3] introduce a Finsler metric whose classifier-induced term “penalises instantaneous motion that transfers probability mass into lineage-inconsistent directions,” making geodesics direction-dependent for single-cell trajectory inference. This is close in spirit to our irreversible directional term, but is a *distance/interpolation* objective in a fixed Finsler metric, with neither a belief-scaled group deformation nor parallel transport of a carried velocity.

Parallel transport in networks. Transport across tangent spaces underlies manifold and mesh convolution [6, 7] and hyperbolic networks [8], but uses the *Levi–Civita* (Riemannian) connection and typically treats path-dependence as a nuisance to be removed via flat connections or smallest-rotation assumptions. We invert this: we use a Finsler *nonlinear (spray) connection* (7) and treat curvature-induced path-dependence as the carrier of meaning.

Holonomy and curvature as representational features. Concurrently (2026), Sevetlidis et al. [9] propose *representation holonomy*—transporting features around a closed loop, with nonzero holonomy revealing hidden curvature—as a robustness diagnostic, essentially our closed-loop check (11)/holonomy used in the opposite direction. Contextual Curvature [10] relate *contextual curvature* of the representation trajectory to next-token entropy in LLMs. The intuition that trajectory curvature/holonomy carries functional meaning is therefore actively and independently emerging; our distinctive claim is constructive (generation *via* an imposed irreversible-Finsler geometry), not diagnostic.

Lie-group deformation and geometry of uncertainty. One-parameter-subgroup / matrix-exponential deformations are standard, convexity-friendly devices in Lie-group networks [11], which is why $M(x) = M_{\text{base}} e^{t(x)A}$ preserves invertibility for free; and several works represent uncertainty geometrically [12]. We are unaware of prior work coupling a *scalar belief field* to a one-parameter-subgroup deformation of a Finsler indicatrix to drive generation.

Summary. Learned Randers metrics [2], direction-dependent learned geodesic cost [3], transport-based networks [6], and holonomy/curvature-as-feature [9, 10] are each precededented or concurrent. What appears unclaimed is their *synthesis*: a belief-scaled, irreversible, convex Finsler metric whose spray-connection curvature transports a velocity so that *route determines context*.

8 Discussion and Limitations

The *geometry* is validated: convex, irreversible, curved, path-dependent, and differentiable end-to-end, and—per the multi-seed study (Table 2)—these hold across every tested seed and dimension rather than in one configuration. The single worked example (holonomy 7.3×10^{-3}) was in fact an unusually weak draw: the mean holonomy at $n = 3$ is 0.25, roughly $35\times$ larger, and both curvature and path-divergence *grow* with dimension, so the effect is not an artifact of low n . Two limitations remain explicit. First, the high-dimension rows are under-seeded ($n = 6$: 2 seeds) because the full spray connection (6)–(7) is $\mathcal{O}(n^4)$ to backpropagate in a naive autograd implementation; a closed-form Randers spray is required to scale the study to $n \geq 8$ at seed count, and the robustness runs use the osculating Levi–Civita transport (validated as path-dependence-equivalent) for speed. Second, the empirical claim is deliberately narrow. On MultiWOZ 2.2 the derived-geodesic model outperforms a parameter-matched attention baseline across three seeds (Section 6.1), but these are small from-scratch models on five closed-set slots, not a comparison with pretrained DST systems, and the mechanism is $\sim 4\times$ slower per step. On next-token language modelling we observe no such advantage: parameter-matched attention and selective state-space baselines match or exceed the mechanism, which is consistent with our reading that the construction pays off where the target is an *accumulated global state* rather than a local continuation. Third, and most importantly, the ablation in Section 6.1 shows the irreversible tilt is inert at the current parameterisation: the gain comes from belief-scaled deformation and multi-velocity geodesic accumulation, not from $F(x, y) \neq F(x, -y)$. Whether irreversibility can be made load-bearing—by increasing $\|c\|$, or by a construction in which the antisymmetric s_{ij} drives rather than perturbs the connection—is the central open question left by this work. Multi-seed statistics at larger scale, additional dialogue corpora, and tasks with explicit directional cost (Zermelo navigation, single-cell trajectory inference) are natural next steps.

9 Summary of Equations

$t(x) = \text{MLP}_\theta(x)$	$A = S + K, [S, K] \neq 0$
$M(x) = M_{\text{base}} e^{t(x)A}$	$F(x, y) = F_0(M(x)^{-1}y), F_0(w) = \sqrt{w^\top w} + c^\top w$
$g_{ij} = \frac{1}{2} \partial_{y^i}^2 \partial_{y^j}^2 F^2$	$G^i = \frac{1}{4} g^{il} (2\partial_j g_{lk} - \partial_l g_{jk}) y^j y^k$
$N^i_j = \partial G^i / \partial y^j$	$\dot{v}^i = -N^i_j(\gamma, v) \dot{\gamma}^j$
$\text{ctx} = (v_0, \dots, v_T)$	$\mathcal{L} = F(x, \pi(c_{\langle t-1 \rangle} c_{\langle t \rangle}^\top))$
$\rho(x) = \max_{\ y\ =1} F(x, y) / F(x, -y)$	

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